Approach to Quantify Eye Movements to Augment Stroke Diagnosis With a Non-Calibrated Eye-Tracker

Mohamed Abul Hassan, Chad M. Aldridge, Yan Zhuang, Xuwang Yin, Timothy McMurry, Gustavo K. Rohde, and Andrew M. Southerland

***Abstract*—Automated eye-tracking technology could enhance diagnosis for many neurological diseases, including stroke. Current literature focuses on gaze estimation through a form of calibration. However, patients with neuro-ocular abnormalities may have difficulty completing a calibration procedure due to inattention or other neurological deficits. Objective: We investigated 1) the need for calibration to measure eye movement symmetry in healthy controls and 2) the potential of eye movement symmetry to distinguish between healthy controls and patients. Methods: We analyzed fixations, smooth pursuits, saccades, and conjugacy measured by a Spearman correlation coefficient and utilized a linear mixed-effects model to estimate the effect of calibration. Results: Healthy participants (n = 18) did not differ in correlations between calibrated and non-calibrated conditions for all tests. The calibration condition did not improve the linear mixed effects model (log-likelihood ratio test p = 0.426) in predicting correlation coefficients. Interestingly, the patient group (n = 17) differed in correlations for the DOT (0.844 [95% CI 0.602, 0.920] vs. 0.98 [95% CI 0.976, 0.985]), H (0.903 [95% CI 0.746, 0.958] vs. 0.979 [95% CI 0.971, 0.986]), and OKN (0.898 [95% CI 0.785, 0.958] vs. 0.993 [95% CI 0.987, 0.996]) tests compared to healthy controls along the x-axis. These differences were not observed along the y-axis. Significance: This study suggests that automated eye tracking can be deployed without calibration to measure eye movement symmetry. It may be a good discriminator between normal and abnormal eye movement symmetry. Validation of these findings in larger populations is required.**

***Index Terms*—Remote health monitoring, conjugate gaze, pupil detection, gaze tracking, eye movements, posterior circulation stroke, acute vestibular syndrome.**

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Mohamed Abul Hassan, Yan Zhuang, Xuwang Yin, and Gustavo K. Rohde are with the Imaging and Data Science Lab, Department of Biomedical Engineering, University of Virginia, USA.

Chad M. Aldridge is with the Department of Neurology, University of Virginia, USA.

Timothy McMurry is with the Department of Public Health Sciences, University of Virginia, USA.

Andrew M. Southerland is with the Department of Public Health Sciences, University of Virginia, Charlottesville, VA 22903 USA (e-mail: as5ef@virginia.edu).

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I. Introduction

POSTERIOR circulatory stroke (PCS) is a type of stroke that often presents with abnormal eye movements. PCS is caused by an impairment of blood flow (ischemia) or bleeding (hemorrhage) in the vascular system that supplies inferior and posterior parts of the brain, including the brainstem, cerebellum, and occipital lobes. These blood vessels, and arteries, map themselves around critical portions of the brain that contain dense collections of neurons known as nuclei [1]. For example, in the brainstem, the cranial nerve VI nucleus functions to move the eye outward horizontally (abduction). A PCS affecting cranial nerve VI's nucleus will paralyze the corresponding eye from abducting itself. If a PCS severely damages the right cranial nerve VI nucleus, then a person will not be able to move that eye to the right, while the left eye remains unaffected by the stroke, as shown in Fig. 1. There are numerous other nuclei that play vital roles in eye movements, coordination, and equilibrium that receive blood supply from the posterior circulation [2].

Clinically, PCS accounts for approximately 25% of Ischemic strokes [3] and is three times [4] more likely to be misdiagnosed than anterior circulatory stroke. Diagnosing symptoms such as abnormal eye movements often requires a trained physician, such as a neurologist or neuro-ophthalmologist. A trained physician would look into deficits of interest such as nystagmus, abnormal visual fixation, and skew deviation [5], [6]. Abnormal eye position and movements due to a PCS can be subtle and difficult to detect by the untrained observer [5], [7].

Commonly used prehospital stroke scales and triage systems do not adequately represent signs and symptoms of PCS, which may also escape detection by cerebral imaging [8]. Additionally, it is challenging for non-experienced providers to differentiate between signs of PCS and acute vestibular syndrome (AVS) at the bedside, particularly in emergency settings [9].

Further, sophisticated procedures such as video-oculography or maneuvers such as the HINTS exam (head impulse, nystagmus, test of skew) [10] are not well generalizable to the prehospital or emergency settings or primary care clinics. Machine learning is used to detect saccades and quick phases in eye movement from a low sampling rate (i.e., 60 Hz) common eye tracker [11], [12], [13] instead of depending on expensive, high sampling rate eye tracking devices. The adaptation of eye tracking with the influence of machine learning could lead to an

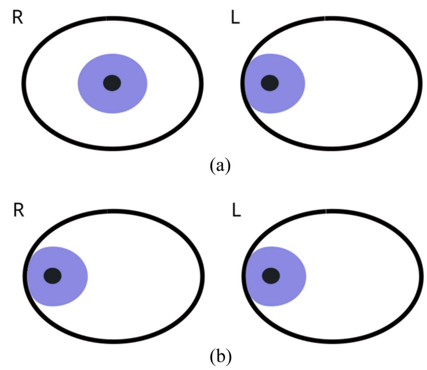

Fig. 1. Illustration of the outward eye movement for (a) pair of eyes affected by PCS that lead to cranial nerve VI palsy. Hence paralyzing the right eye from abducting itself. (b) the normal function of cranial nerve VI. (a) Cranial nerve VI function-PCS. (b) Cranial nerve VI function-Normal.

eye-tracking screening tool that could be implemented in most clinics and prehospital settings. This could lead to more accurate diagnosis and rapid treatment in cases of PCS, resulting in better patient outcomes.

The most significant barrier to using eye tracking for stroke-related neuro-ocular deficit detection is the issue surrounding the calibration procedure required by eye-tracking devices. For commercial binocular eye trackers, the eyes need to move and fixate their gaze symmetrically on the dots in the screen to reach the accuracy specified by the manufacturer. This is difficult for patients affected by stroke-related neuro-ocular deficits. Furthermore, several studies investigating the application of eye-tracking to various neurological diseases [13], [14], [15], [16], [17], [18], [19], [20] had to exclude patients that cannot complete the calibration procedure fully or at least partially. This reduces the generalizability of commercially available trackers for practical use in a clinical setting. To address this issue, we performed a feasibility study to quantify if the calibration procedure is necessary to measure a fundamental neuro-ocular characteristic, eye conjugacy [21].

Conjugacy refers to the angular alignment of the eyes during binocular movements, which is examined by various clinical tests to diagnose ocular palsies, vertical skew, and other forms of misalignment. There are two primary aims of this feasibility study: 1) to determine the difference in eye conjugacy of calibrated and non-calibrated measures in healthy controls, and 2) to investigate the potential for a non-calibrated measurement of eye conjugacy to discriminate between patients with neuro-ocular deficits and healthy controls for future development of a machine learning tool.

The descriptions and outcomes of our feasibility study are presented as follows: Section II presents a brief description of the related work that has used eye tracking for clinical applications. Section III investigates the impact of calibration on POG and derives a mathematical model based on the physiological properties of the eye and working principles of gaze estimation. The experimental setup of the feasibility study is presented in section IV. Results and discussion for the feasibility study are presented in section V, and section VI provides an analysis of the study with context to stroke examination. The conclusions and limitations of the study are presented in Sections VII and VIII.

II. Related Work

Eye tracking is widely used for multiple clinical applications, particularly neurologic and psychiatric diseases. For example, researchers have used eye tracking to quantify the mental state of Alzheimer's patients. The researchers used a mobile device-based eye tracker to perform cognitive assessments [22]. The study consisted of 15 participants diagnosed with mild Alzheimer's. The study reported that all 15 participants completed a calibration procedure before undergoing the assessment.

Deng et al. [23] used eye-tracking data to discriminate participants diagnosed with attention-deficit/hyperactivity disorder from healthy controls. The study adopted a free-viewing stimulus to invoke fixation, saccades, and scan-path-like eye movement behavior. The eye movements were captured using an infrared video-based eye tracker and recorded as monocular eye tracking data. The study used 1246 participants, of which 232 were diagnosed with attention-deficit/hyperactivity disorder, and 152 were unaffected controls. The study did not report on the use of calibration procedures.

Zhang et al. [24] investigated eye movement abnormalities to identify psychosis in individuals at clinically high risk. The study involved 178 participants, of which 108 were the clinically high-risk population, and 70 were control participants. Eye tracking was used for fixation and free-viewing tasks. The eye-tracker acquired gaze points while the participant's head was kept still by placing it on a chin rest. The study protocol required a 9-point calibration procedure to acquire gaze points.

Kapoula et al. [25] performed gaze estimation by detecting abnormal eye movements such as optokinetic-nystagmus (OKN) and smooth pursuit (SP) for five participants suffering from somatic tinnitus. Here the researchers performed a custom calibration using a linear fitting function to align the gaze data with the actual targets.

Puuganti et al. [11], [12] investigated saccades detection during nystagmus using a wearable eye tracker while performing the Dix-Hallpike maneuver. The detection algorithm was tested on a dataset of 103 patients performing the Dix-Hallpike maneuver. The wearable eye tracker was calibrated for all the cases using a laser target projected from the goggles. Researchers have used eye trackers to understand motion blur caused by nystagmus [26], and attention deficit [27] and included calibration as a vital step in the study workflow.

Kumar et al. [13] used an eye tracker to measure abnormal eye movements of eight stroke survivors during the rehabilitation process. The study involved eight stroke survivors and eight healthy participants. Before performing a set of eye-tracking tasks, the participants had to complete a calibration procedure or be excluded from the study.

Bijvank et al. [18], [19] used eye tracking to detect fixation-related abnormal eye movements caused by multiple sclerosis. The study included 213 patients diagnosed with multiple sclerosis and 57 healthy controls. A custom calibration procedure

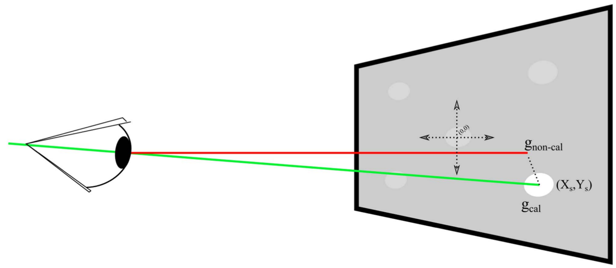

Fig. 2. Illustration of the calibration process.

based on four fixation locations was used before performing the eye-tracking tasks. The study protocol required the participants to successfully complete the calibration procedure or be excluded from the study. Thirteen participants with multiple sclerosis were excluded from the study due to calibration failure.

Jinsook et al. [17] performed a study investigating the ocular motor function of patients suffering from cerebral ataxia. The study involved detecting saccades using an eye tracker while performing a simple reading task. The study involved 11 patients and 11 controls. However, none of the patients presented with dysconjugate eye movements. All participants were required to complete a nine-point calibration procedure. This led to the exclusion of recordings with poor calibration accuracy.

III. Understanding Calibration

The point of Gaze (POG) is defined as the intersection of the eye's visual axis with the screen. Eye trackers use a gaze calibration procedure to estimate the intersection of the eye's visual axis with the scene (see Fig. 2). Essentially the process of calibration uses a minimization function to reduce the error between the estimated gaze point $\mathbf{g}_{non-cal}$ and the calibration point/target in the screen $\mathbf{g}_{cal}$ (see Eq. (1)):

$$f : \mathbf{g}_{non-cal} \rightarrow \mathbf{g}_{cal}, \tag{1}$$

With $\mathbf{g}_{cal}$ and $\mathbf{g}_{non-cal}$ both points on the screen, and with the underlying assumption that the subject is indeed focusing their gaze on the screen point $\mathbf{g}_{cal}$. Generally speaking [28], [29], [30], the function f is defined as

$$f(\mathbf{g}) = \mathbf{t} + A\mathbf{g} + \sum_{i=1}^{M} \mathbf{c}_i \phi_i(\mathbf{g}) \tag{2}$$

where $\mathbf{t}$ is a translation vector, A a 2x2 matrix encompassing scaling, shear, and rotation, and $\mathbf{c}_i$ coefficients for nonlinear basis functions (e.g., polynomials or radial basis functions) to account for any non-linearities present in the system. In this formulation, the calibration model parameters are $\mathbf{t}$, A, and $\mathbf{c}_i$, $i = 1, \ldots, M$.

With knowledge of a mathematical model and measured points, the goal of the calibration procedure is to obtain many corresponding measurements $\mathbf{g}^k_{cal}, \mathbf{g}^k_{non-cal}, k = 1, \ldots, N$, and then finding model parameters $\mathbf{t}$, A, and $\mathbf{c}_i$ that minimize some measure of error between the calibrated and non-calibrated points. One popular error function is least squares, in which case the optimization problem can be defined as:

$$\min_{\mathbf{t}, A, and \mathbf{c}_i} \Psi(\mathbf{t}, A, \mathbf{c}_i) \tag{3}$$

with

$$\Psi(\mathbf{t}, A, \mathbf{c}_i) = \frac{1}{N} \sum_{k=1}^{N} \| f(\mathbf{g}_{non-cal}) - \mathbf{g}_{cal} \|^2. \tag{4}$$

Understanding the above, it becomes clear that whichever procedure is being used by a particular device manufacturer to obtain $\mathbf{g}_{non-cal}$, uncalibrated points tend to be a 'deformed' version of $\mathbf{g}_{cal}$. In cases where the nonlinear component of the model expressed in (4) is negligible, the points will be translated and rotated/scaled/sheared versions of the actual points (assuming that the patient is actually looking exactly at the stimulus point presented in the screen). In such cases, there are a few observations with which uncalibrated data can be used for obtaining quantitative information to differentiate normal vs. abnormal populations:

- *Accuracy:* Calibration is required to reliably measure the point of gaze coordinates with respect to the screen coordinates.
- *Agreement between left and right eye:* Calibration is not required to quantify relative movement between the left and right. As expressed in (4), if the nonlinear portion of the model is deemed negligible, the correlation between uncalibrated left and right eye movements will be linear [31]. This is due to the fact that the remaining affine portion of the model in (2), by definition, does not affect linear correlations.
- *Discriminant analysis:* often, pattern recognition methods are used on a collection of gaze points estimated from a subject during an examination procedure to 'learn' differences between the normal and the abnormal populations. Therefore, pattern recognition methods can be invariant concerning affine transformations. Hence, discriminant analysis is possible.

IV. Experiment Setup

The experimental protocol was approved by the University of Virginia's Institutional Review Board. As such, the protocol complies with all national ethical research standards and is in accordance with the Declaration of Helsinki. Written informed consent was obtained prior to subject enrollment and testing of hospitalized patients and healthy participants. Patients and healthy participants had to be adults aged 18 years or older. Patient enrollment criteria consisted of 1) presentation with an acute onset of continuous dizziness lasting at least one hour and within one week of onset and 2) ability to provide informed consent. Dizziness included lightheadedness, vertigo or disequilibrium, and head motion intolerance with spontaneous or gaze-evoked nystagmus. Patients were excluded if they: 1) were unable to provide consent, 2) had previous vestibular dysfunction, pseudo-vertigo or vestibular migraine, severe cognitive dysfunction, or other prior injuries that prevented participation, 3) had acute drug or alcohol intoxication or head trauma, or 4) symptoms abated within 24 hours of onset or

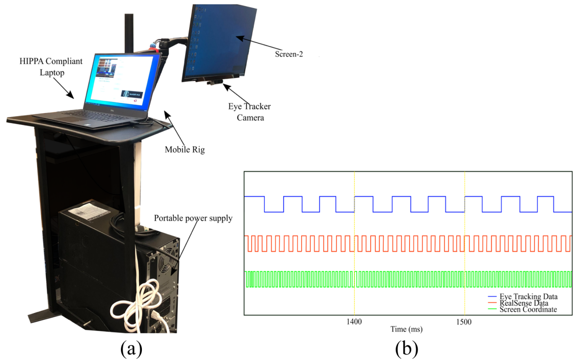

Fig. 3. Illustration of the Rolling Apparatus to Detect Impairment of the Eyes - RoADIE and the trigger-based real-time synchronization process used during the clinical tests. (a) Illustration of the main hardware components for RoADIE used for data acquisition, (b) Trigger-based synchronization of the eye-tracker, RealSense, and the screen coordinates from the stimuli.

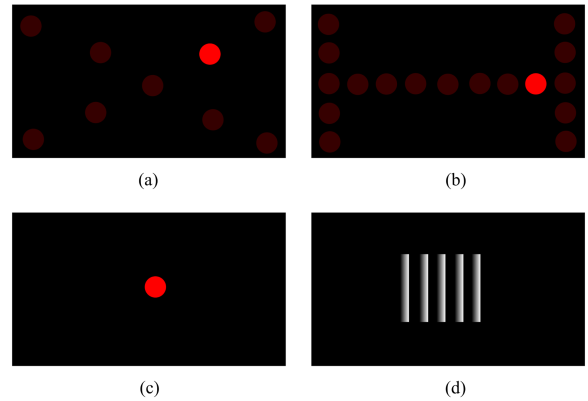

Fig. 4. Illustration of the NeuroEye examination. (a) represents the Dot-Test for the Neuro-eye examination. The red dot with the higher contrast represents the current position of the dot, and the red dot with the lower contrast represents the past, or future position of the dot (b) represents the H-Test for the Neuro-eye examination. The red dot with the higher contrast represents the current position of the dot, and the red dot with the lower contrast represents the past or future position of the dot (c) Illustration of the stationary dot used for HO-Test (d) Illustration of the OKN-Test.

prior to consent. To be enrolled, healthy participants must have had no history of vestibular or ocular motor deficits. The experimental setup of our feasibility study to assess the ability of non-calibrated gaze tracking to detect ocular motor deficits had two main components: 1) RoADIE (Rolling Apparatus to Detect Impairment of the Eyes), and 2) NeuroEye, a neurology eye examination comprised of four different tests investigating various ocular motor functions.

A. RoADIE

RoADIE is a mobile rig developed to acquire gaze data while the participant undergoes the NeuroEye examinations. RoADIE is equipped with a HIPPA-compliant computer to run our custom-built data acquisition software and store data. The peripheral devices (i.e., screen stimuli, Eye tracker, and RealSense) were initiated with global trigger and set to acquire data at their native sampling frequency. The Tobii Pro Fusion Eye Tracker estimated gaze at 120 Hz [32], and the RealSense camera acquired images at 30 Hz [33] while observing the stimuli of the NeuroEye exam (See Fig. 3(a)). As a post-processing step, the eye tracking and RealSense data were aligned with the screen coordinates at 100 ms intervals as shown in Fig. 3(b)).

B. NeuroEye

The NeuroEye examination comprises four different tests, investigating various motor functionalities of the eye. The four tests are the Dot-Test, H-Test, OKN-Test, and HO Test. They are computer adaptions of standard bedside ocular motor tests. The Dot Test is designed to assess the quality of eye coordination of a volitional saccade. Similar to a clinician, the participant is asked to shift their gaze from dot to dot as they appear (see Fig. 4(a)). A clinician looks for the eyes to suddenly move to the next visual target and stop accurately at its destination, which the Dot-Test recreates. Abnormal signs include the eyes stopping short or too far away from the target, known as under or over-shooting, respectively, or the eyes do not initiate a saccade.

The H Test assesses the quality of eye movement at a constant slow pace as the eyes track a visual target in an “H” pattern that encompasses the four cardinal ocular directions and quadrants, mirroring a bedside examiner having a patient follow the examiner’s finger with their eyes. The “H” pattern (see Fig. 4(b)) ensures that both eyes track the target in all cardinal directions to their end ranges. Abnormal signs consist of the eyes lacking motion or its initiation in any direction or the eyes utilizing small saccades to catch up to the moving visual target (i.e., abnormal smooth pursuit).

The Head Orientation (HO) Test is designed using a stationary dot that would appear at the center of the screen (see Fig. 4(c)) and requires the patient to be able to tilt the head to the left and right in alternate fashion. Head tilting exacerbates a vertical misalignment, mainly due to an ocular tilt reaction (OTR), as the head tilts toward the superiorly positioned eye relative to the other. Conversely, the head tilting toward the inferiorly positioned eye will reduce the patient’s experience of diplopia as the eyes become level with each other. In patients with (OTR), one expects the patient’s initial head tilt to be biased toward one side. Thus, the amount of movement in eye alignment due to head tilting may be assumed to be symmetric in healthy participants and more asymmetric in patients suffering a lesion affecting gravitational perception from the utricle and vertical canal structures of the inner ear.

The optokinetic nystagmus (OKN) test measures the participant’s ability to switch from smooth pursuit to a saccade in order to fixate on the next visual target after the first target disappears. The visual stimulus typically comprises vertical bars with high contrast to the background. The bars move at a quick and constant pace from right to left (see Fig. 4(d)). This is done for each opposite direction. OKN typically remains preserved in individuals with occipital lobe infarcts, although impairment of a visual field may limit the amplitude of saccadic fixation. Asymmetry in the performance of the eyes or poor ability to

TABLE I
MEAN CORRELATION AND 95% CIS BY CALIBRATION CONDITION AND GROUP FOR EACH TEST FROM 10,000 BOOTSTRAPPED PARTICIPANT-LEVEL SPEARMAN CORRELATION COEFFICIENTS BETWEEN THE LEFT AND RIGHT EYES FOR EACH AXIS.

Axis	Test	Calibration Mean	Calibration 95% CI	Non-Calibrated Mean	Non-Calibrated 95% CI	Difference Mean	Difference 95% CI
Healthy X	DOT	0.982	0.974, 0.988	0.980	0.976, 0.985	0.002	-0.006, 0.007
	H	0.980	0.974, 0.986	0.979	0.971, 0.986	0.001	-0.006, 0.009
	HO	0.525	0.349, 0.686	0.561	0.371, 0.727	-0.037	-0.139, 0.052
	OKN	0.993	0.988, 0.996	0.993	0.987, 0.996	0	-0.006, 0.007
Patient X	DOT			0.844*	0.602, 0.920		
	H			0.903*	0.746, 0.958		
	HO			0.419	0.276, 0.553		
	OKN			0.898*	0.785, 0.958		
Healthy Y	DOT	0.953	0.91, 0.981	0.948	0.924, 0.969	0.005	-0.044, 0.044
	H	0.846	0.751, 0.923	0.887	0.869, 0.907	-0.041	-0.129, 0.030
	HO	-0.050	-0.306, 0.210	-0.070	-0.260, 0.142	0.020	-0.171, 0.233
	OKN	0.785	0.695, 0.862	0.706	0.604, 0.799	0.079	-0.052, 0.212
Patient Y	DOT			0.922	0.869, 0.949		
	H			0.886	0.834, 0.919		
	HO			0.303	0.104, 0.464		
	OKN			0.581	0.282, 0.713		

generate saccades during the OKN test may suggest damage to the ocular, brainstem, or cerebellar nuclei or tracts.

C. Statistical Plan

Statistical analyses were conducted with R (v4.0.3, R Foundation for Statistical Computing, Vienna, Austria). For all four tests, we assumed that a healthy ocular motor system consists of highly correlated position changes between the left and right eyes for the X and Y axis. Additionally, we assumed that the relative rate of change between both eyes is constant, which results in a linear relationship [34]. This assumption reflects the normal physiology of human eye movement to have "perfect coordination," which is shared among all primate species [35]. Therefore, we applied Spearman's rank correlation on raw eye tracking data to quantify the strength of the linear relationship between the two eyes [31]. The correlation coefficient determines the relationship of a linear slope with a value from -1 to 1. Zero means that there is no consistent relationship of change between two eye movements. A value of one signifies a perfect relationship of position change between two eye movements.

We chose Spearman's coefficient over Pearson's due to the non-Gaussian distribution of coordinates. Because of the small sample of participants, we performed 10,000 bootstraps of the Spearman's coefficient of the participants to estimate the average correlation coefficient and its 95% confidence intervals for each axis, calibration condition, and test. This ensures that the sample of participant-level correlation coefficients is independent. Therefore we repeated the approach for the difference in correlation coefficients and its 95% confidence interval between calibration conditions. We defined the statistical significance of the difference in correlation coefficients between calibrations conditions as the bootstrapped 95% confidence interval not containing zero. Furthermore, if the 95% confidence intervals of the correlation coefficients of patients did not overlap the respective confidence intervals of the non-calibrated correlation coefficients of the healthy participants, then the groups were considered significantly different. Table I shows these results.

To estimate the overall effect of calibration on the eye-tracker's ability to measure the symmetry of eye movement, a linear mixed effects model was implemented with the "lme4" [36] package to account for the repeated measurement among participants and their random effects seen in Fig. 5(b). Participant-level Spearman correlations were the response variable with the following covariates: calibration condition, axis (X/Y), and NeuroEye exam as fixed effects, with the participant as a random effect. Participants as a random effect allow the intercept for the fixed effects to vary for each participant. Fig. 1 demonstrates the need for participants to have their own intercepts. Prior to fitting the model, the Spearman correlations, the response variable, received a Fisher's Z transformation to achieve a normal distribution. Additionally, we implemented weights to account for the different number of observations from each test as follows: 350 (DOT), 220 (H), 350 (HO), and 350 (OKN). The weights also improved homoscedasticity. The linear mixed-effects model showed an even distribution of residuals and fitted values. To test the significance of the calibration fixed effect, we performed a log-likelihood ratio test between the previously described model with a simpler model without the calibration covariate.

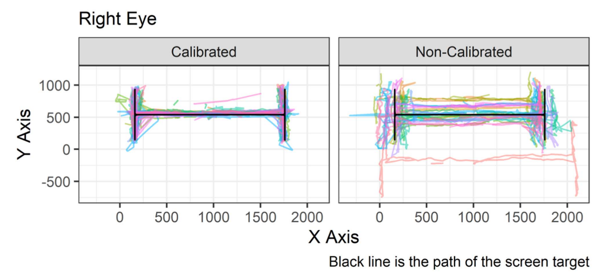

Fig. 5. Calibration and non-calibration path plot to visualize the performance of the H Test. The line colors represent each participant's test.

Equation (5) is the formula of the mixed-effects model where y is a $N \times 1$ response variable vector, X is a $N \times p$ matrix of p predictor variables, β is a $p \times 1$ vector of fixed effects, Z is a $N \times qJ$ design matrix for q random effects and J groups, u is a $qJ \times 1$ vector of q random effects for J groups, and lastly, ϵ is a $N \times 1$ vector of the residuals.

$$y = X\beta + Zu + \epsilon \tag{5}$$

The predictor variables, p, consist of the following indicator variables: calibration (0/1), y-axis (0/1), H test (0/1), HO test (0/1), and OKN test (0/1). The non-calibration condition, x-axis, and DOT test are references for the corresponding calibration, axis, and NeuroEye fixed effects. The random effect, q, represents the random intercept of each healthy participant. Because the participants are not nested into groups, J is a vector of 1's. Thus the design matrix Z becomes a matrix with 288 rows and 18 columns. Each healthy participant has 16 observations (calibration condition (n=2), axis (n=2), and test (n=4)) and a specific column indicating which observation corresponds to which participant. The simpler model removes the calibration indicator variable, which reduces the number of predictor variables, p, from 5 to 4.

V. Results

Nineteen healthy participants enrolled in the calibration study. The mean age (range) is 40 (25-62) years and 79% female. The racial distribution of the healthy participants is 81% White, 14% Asian, 5% American Indian, or Alaska Native. Comparatively, twenty-three patients were screened and of those, 17 enrolled. The patients consisted of 9 (53%) females with and mean age (range) of 61.8 (19-96) years. The patients had a racial distribution of 14 (82%) White, 2 (12%) Black, and 1 (6%) Other.

Patient enrollment and data collection were completed within a median of 3 days ranging from 0 to 37 days of symptom onset. Acute ischemic stroke accounted for 13 patients with the rest having a diagnosis of Acute Vestibular Syndrome of suspected central origin (n=2), Multiple Sclerosis (n=1), and Vestibular Neuritis (n=1). Diplopia, or double vision, was the primary presenting symptom in six patients, with nystagmus or vertigo being the presenting characteristic in 7 patients. Head impulse testing showed that 7 patients had normal responses, while four had an impaired vestibular-ocular reflex. The head impulse test was not performed for 6 patients. None of the patients had acute hearing loss. Only four patients had impaired smooth pursuit.

One of the healthy participants did not follow instructions during the experiment and moved out of the required testing position, which prevented eye gaze capture. Therefore, this participant was excluded from the study. None of the patients were able to complete the Tobii eye-tracker calibration procedure. Thus, there are no calibrated correlation measures for patients as seen in Table. I.[1]

There were no significant differences in the correlation between calibrated and non-calibrated conditions of controls. As Table I shows, differences between the correlation of calibrated and non-calibrated procedures were consistently near zero, with their 95% confidence intervals overlapping zero. The calibration parameter in the linear mixed effects model failed to achieve significance from the log-likelihood ratio test (p-value 0.426, $\chi^2 0.635$ with 1 degree of freedom) from the simpler model without the calibration parameter. Both calibration conditions exhibited similar variance, as shown by the range of the 95% confidence intervals. For example, the DOT test on the X axis had a 95% CI range of 0.014 for calibrated and 0.009 for non-calibrated. The variance was different among the NeuroEye tests as well. The HO test consistently had the greatest range of 95% CI compared to the rest. The different variances among tests were also consistent between calibrated and non-calibrated conditions. Visually, the structure of gaze coordinates over time is similar between calibrated and non-calibrated conditions, demonstrated by Fig. 5 showing the "H" structure or path that the eyes track by following the visual target during the H test. Even though the path that the eyes take has visual differences between conditions, Fig. 6 shows that the linear relationship between the eyes remains intact.

[1] There were no significant differences between calibration conditions of healthy participants because all 95% CIs of mean differences encompassed zero. Significant differences between non-calibrated correlation coefficients of healthy participants and patients are marked with a *.

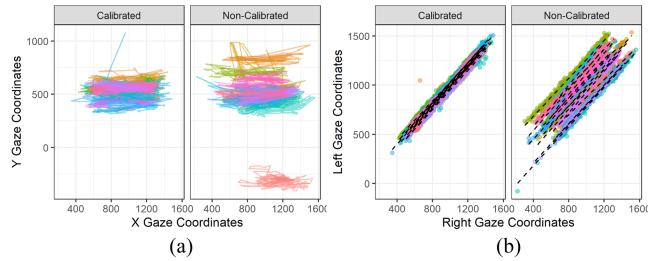

Fig. 6. Analysis for the OKN test. (a) The path of the right eye over time for each participant by calibration condition during the OKN test. (b) The correlation of the right and left eye on the X axis by calibration condition during the OKN test. A linear regression line overlays each participant's scatter plot.

Compared to the ocular behavior observed in healthy participants, patients had generally worse correlation coefficients and larger 95% CI ranges; see Table I. For the X axis, only the HO test failed to achieve a significant difference compared to the non-calibrated correlation coefficient of the participants. Referring to the DOT test on the X axis again, patients had a significantly smaller correlation coefficient mean with a 95% CI range of 0.29, which is 3.22 times the range of the non-calibrated correlation coefficient of their control counterparts. In contrast to the X-axis, the Y-axis correlation coefficients of the patients had less variance. The DOT test had a 95% CI range of 0.176 on the Y axis, which is 0.61 times less than the X axis for patients.

Interestingly, Fig. 7 shows the most significant variation of gaze correlation between patients and healthy participants. One can see noticeably different slope angles and clustering of points off the main slope for patients compared to the non-calibrated gaze coordinates of the healthy participants. Additionally, there is much more noise in the structure of the gaze path during the H test, as seen in Fig. 8.

VI. Discussion

The objectives of this study were to 1) test whether an off-the-shelf eye tracker could measure eye conjugacy without a calibration procedure in healthy participants and 2) determine if eye conjugacy measured by a correlation coefficient can discriminate between groups that have normal and abnormal eye movements. The preliminary results suggest that a calibration procedure is not necessary to observe and quantify tandem and

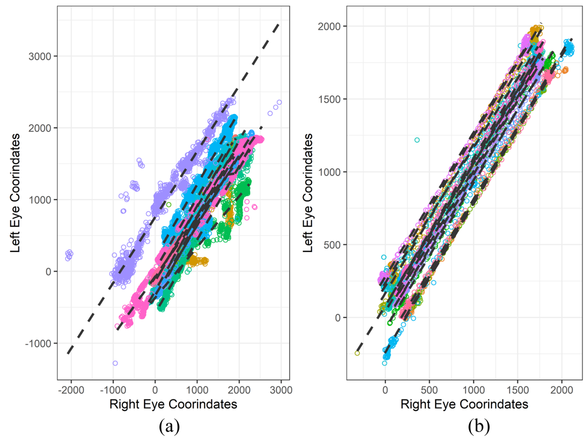

Fig. 7. Relationship of movement between the right and left eye during the H test for Patients and participants without calibration. (a) refers to patients and (b) refers to healthy participants. The dotted lines are robust linear regression overlays for each subject. Each color represents an individual subject.

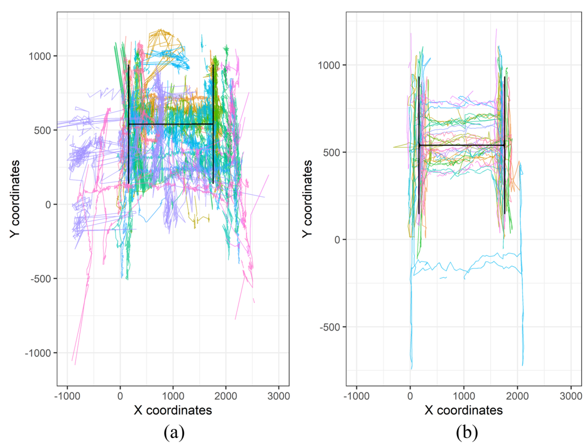

Fig. 8. Path of the right eye during the H test for patients and participants without calibration. (a) refers to patients and (b) refers to healthy participants. Each color represents an individual subject.

correlative eye movements that occur during visual tracking and gaze shifting to various targets.

Table I and Fig. 5 support this assertion. However, when the goal is to achieve accurate and precise gaze estimates at the moment in time, then calibration must be performed. Refer to Fig. 5 to see the transformed gaze estimates from the calibration procedure compared to the non-calibrated gaze estimates of the healthy participants. As described in (4), the calibration condition appears to scale and center the estimated eye positions close to the actual screen coordinates. Still, it does not affect the general path shape created by eye movements.

When the approach of measuring eye conjugacy is applied to patients with ocular motor deficits, it becomes clear that eye movement correlation has the potential to distinguish between patients and healthy participants, in other words, normal physiology versus pathology. The mean gaze correlation for both the X and Y axis are worse for the DOT, H, and OKN tests in patients compared to healthy participants. Still, only the X-axis correlation coefficients were significantly different. This is possibly due to the more significant variance observed in Y axis versus X-axis mean correlation coefficients in healthy participants, which makes discrimination between patients and healthy participants more difficult, requiring larger sample sizes for adequate statistical power.

Another possibility is a clustering of ocular motor deficits affecting horizontal eye movements compared to vertical movements, such as a cranial nerve six palsy, resulting in a worse correlation along the X axis compared to the Y axis. For example, Table I shows that the H test gaze correlation coefficient is similar in both calibration conditions for healthy participants. However, the OKN test has a much lower correlation coefficient mean on the Y axis compared to healthy participants and a high 95% CI range of 0.431 compared to the equivalent of the non-calibrated healthy participants with a 95% CI range of 0.195. The patients have 2.2 times the variance as healthy participants, which supports the first possibility that this study lacked sufficient power to detect significant differences along the Y axis between patients and healthy participants.

Concerning feasibility, it is important to note that none of the patients could complete the Tobii eye tracker calibration procedure. Results for the patient participants reflect real-world ocular motor deficits and diagnoses traditionally excluded from previous studies that included various cohorts with neurological diseases [13], [18], [19]. The patient cohort in this study would require a custom calibration procedure requiring an arbitrary completion threshold to calibrate successfully. Other studies have utilized this approach but still have had to exclude clinically relevant patient populations [18], [37]. For this study, a custom calibration procedure is out of scope as the study's goal was to demonstrate the feasibility of off-the-shelf eye-tracking technology for clinical use. However, this raises the issue that excluding patients who cannot complete a calibration procedure will meaningfully limit the generalizability of results as the studied cohorts do not reflect the true population and heterogeneity of patients suffering from ocular motor deficits in the real world, such as those presenting with posterior circulation stroke.

VII. Conclusion

This study demonstrates that a commercially available eye tracker can measure ocular movement and symmetry with and without calibration in healthy participants. Additionally, gaze correlation as a measure of eye conjugacy can potentially discriminate between healthy and pathological eye movements. Our preliminary findings show that implementing off-the-shelf eye-tracking technology is feasible in healthy adults and those with neurological injuries. This is further strengthened by the fact that none of the enrolled patients could complete the calibration procedure. This is concerning as current eye-tracking research excludes patients unable to perform a calibration procedure. This opens the door to utilizing a machine learning tool to learn relevant features of normal and abnormal eye movements. Machine learning techniques could potentially be used to assess and diagnose neuro-ocular abnormalities without needing a calibration procedure, so long as the analysis is based on relative and correlative eye measurements. Larger sample sizes are necessary to determine this approach's true discriminatory ability prior to testing the technology in a clinical observational trial, particularly to employ more formal machine learning-based diagnosis of abnormal eye movements in patients presenting with posterior circulation stroke and other neuro-ophthalmological disorders.

VIII. Limitations

There are notable limitations to this study. First, as a preliminary proof-of-concept, our study included small sample sizes and lacked race/ethnic diversity. The small sample size of patients does not account for the full range of ocular motor deficits and their respective pathologies in real-world clinical settings. The full extent of a gaze correlation coefficient to discriminate between normal and abnormal eye movements will require further research with larger and more diverse sample sizes and a greater range of symptom presentation of ocular motor deficits. Even though the sample size of healthy participants is small, the study demonstrated sufficient power to detect differences between calibration conditions as the observations within the healthy participants were paired with much lower variance in general compared to patients.

Despite the small sample size, it is reasonable to assume that the healthy participants likely represented the distribution of normal extraocular movements seen in a healthy adult population. For the HO test specifically, this test demonstrated poor gaze correlation coefficients in patients and healthy participants and much higher variance compared to the other three NeuroEye tests. Currently, the HO test does not perform well with eye-tracking technology. Additional research is required to investigate this problem as ocular movement, and potential alignment in response to head-tilting is an important clinical measure to diagnose neurological diseases affecting eye conjugacy, such as stroke, multiple sclerosis, and other brainstem disorders.

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